METRIC
A-A-59599A
23 January 2001
SUPERSEDING
A-A-59599
27 September 2000

COMMERCIAL ITEM DESCRIPTION

BEARING, BALL, ANNULAR, DOUBLE ROW, RADIAL, INTERNAL SELF-ALIGNING, DIMENSION SERIES 22

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers metric, double row, radial, internal self-aligning, annular ball bearings for general-purpose use.

2. CLASSIFICATION. The ball bearings shall be classified by the class, sizes, cage materials, shield and snap ring configurations, precision tolerances, and lubricant, preservative, and grease fill requirements listed below:

Class 6 - dimension series 22

Size - bearing dimensions (see table I)

Cage materials (see table II)

Bore designs (see table III)

Precision tolerances (see table IV)

Lubricant, preservative, and grease fill requirements (see table V)

3. SALIENT CHARACTERISTICS

3.1 Dimensions. Bearing boundary dimensions (and dynamic load ratings, see 3.5) shall conform to the requirements specified in table I for each of the coded bearing sizes. The listed dimensions conform to the requirements specified for the listed bearing sizes from dimension series 22 in the American National Standards Institute/American Bearing Manufacturers Association (ANSI/ABMA) Standard 20, "Radial Bearings of Ball, Cylindrical Roller, and Spherical Roller Types, Metric Design" (DoD adopted). The bearing size shall be as specified in

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3110

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.

the acquisition order (see 7.4(b)). For unlisted bearing sizes, the associated dimensional and dynamic load rating requirements should also be specified in the acquisition order.

TABLE I. Dimensional requirements (ABMA series 22).

Bearing size code	Bore Diameter (mm)	Outside diameter (mm)	Width (mm)	Chamfer r/min (mm)	Minimum shaft shoulder diameter* (mm)	Dynamic load rating (minimum) (lb)
00	10	30	14	0.6	15	1,640
01	12	32	14	0.6	17	2,000
02	15	35	14	0.6	20	2,080
03	17	40	16	0.6	22	2,550
04	20	47	18	1.0	26	2,830
05	25	52	18	1.0	31	2,760
06	30	62	20	1.0	36	3,400
07	35	72	23	1.1	42	4,850
08	40	80	23	1.1	47	5,000
09	45	85	23	1.1	52	5,200
10	50	90	23	1.1	57	5,200
11	55	100	25	1.5	63.5	5,950
12	60	110	28	1.5	68.5	7,644
13	65	120	31	1.5	73.5	9,750
14	70	125	31	1.5	78.5	9,800
15	75	130	31	1.5	83.5	9,937
16	80	140	33	2.0	90	10,900
17	85	150	36	2.0	95	13,100
18	90	160	40	2.0	100	15,220
19	95	170	43	2.1	107	18,592
20	100	180	46	2.1	112	18,187
21	105	190	50	2.1	117	21,334
22	110	200	53	2.1	122	26,977
24	120	215	58	2.1	132	23,130
26	130	230	64	3.0	144	26,870
28	140	250	68	3.0	154	28,640
30	150	270	73	3.0	164	33,485

* Listed for reference purposes only. Shoulder height shall be determined to provide sufficient clearance for the direct application of bearing removal force against the bearing inner ring. If the required minimum clearance is not available, an alternative non-destructive bearing removal capability shall be provided.

3.2 Materials

3.2.1 Rings. The bearing ring material shall be chromium-alloy steel 52100 (UNS G52986) as specified in the American Society for Testing and Materials (ASTM) A 295, "Standard Specification for High-Carbon and Roller Bearing Steel" (DoD adopted). The finished rings

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.

shall not exceed the associated billet material inclusion rating also specified in ASTM A 295. Ring hardness shall be at least 58 HRC as defined in ASTM E 18, "Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials" (DoD adopted). The grain size of the ring material shall be determined in accordance with ASTM E 112, "Standard Methods for Determining Average Grain Size" (DoD adopted).

3.2.2 Balls. The bearing ball material shall be chromium-alloy steel 52100 (UNS G52986) as specified in ASTM A 295. The finished balls shall not exceed the associated billet material inclusion rating, which is also specified in ASTM A 295. Balls shall be through-hardened no less than 60 HRC and no more than 67 HRC as defined in ASTM E 18. The grain size of the ball material shall be determined in accordance with ASTM E 112.

3.2.3 Cage. The bearing cage material shall be impervious to deterioration from any lubricant, preservative, solvent, or other chemical substance expected to contact the bearing during normal use or storage. Similarly, the material shall not cause any chemical deterioration of any other bearing component. Metallic and non-metallic cages shall meet the same bearing performance requirements. Cage materials shall operate from -65 to 230 °F (-54 to 110 °C). Unless otherwise specified in the acquisition order, the cage material shall be one of the optional coded types listed in table II (see 7.4(c)).

TABLE II. Cage materials.

Code[1]	Material type
J	Pressed steel
M	Machined bronze or brass
P	Molded plastic (nylon 66 or equal)
T	Machined non-metallic (phenolic)
Y	Pressed brass
X	Other (specify in acquisition order)

[1] No significant change from FF-B-171/13.

3.3 Bore design. All self-aligning bearings shall have either a cylindrical or tapered bores (taper 1:12) and shall be identified by one of the codes listed in table III. The bore design shall be specified in the acquisition order (see 7.4(d)).

TABLE III. Bore designs.

Code[1]	Configuration option
C	Cylindrical bore (no shields or seals)
T	Tapered bore (no shields or seals)

[1] No significant change from FF-B-171/13.

3.4 Precision tolerance. The bearing precision tolerance level shall conform to Annular Bearing Engineers Committee (ABEC) class ABEC-1, ABEC-3, ABEC-5, or ABEC-7 as defined in ANSI/ABMA Standard 20. The tolerance class and associated radial internal clearance shall be one of the coded options listed in table IV (see 7.4(e)).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.

3.4.1 Bearing precision tolerance levels and internal clearances may be affected by the addition of shields and/or seals. Manufacturers/suppliers should be consulted for availability of desired precision tolerances when shields and/or seals are required.

TABLE IV. Precision tolerance requirements.

Code	Tolerance class	Radial internal clearance
A	ABEC-1	Symbol 2 (less than normal)
B		Symbol 0 (normal)
C		Symbol 3 (greater than normal)
D		Symbol 4 (greater than symbol 3)
E	ABEC-3	Symbol 2 (less than normal)
F		Symbol 0 (normal)
G		Symbol 3 (greater than normal)
H		Symbol 4 (greater than symbol 3)
J	ABEC-5	Symbol 2 (less than normal)
K		Symbol 0 (normal)
L		Symbol 3 (greater than normal)
M		Symbol 4 (greater than symbol 3)
N	ABEC-7	Symbol 2 (less than normal)
P		Symbol 0 (normal)
R		Symbol 3 (greater than normal)
S		Symbol 4 (greater than symbol 3)

3.5 Dynamic load rating. The listed ratings in table I conform to the requirements specified in ANSI/ABMA Standard 9, "Load Ratings and Fatigue Life for Ball Bearings" (DoD adopted).

3.6 Lubrication, preservation, and grease fill requirements. Unless otherwise specified in the acquisition order, the bearing lubricant or preservative and grease fill requirement shall be as selected from the approved coded options listed in table V (see 7.4(f)). The percentage of fill is based on the internal empty space of an assembled bearing.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Unless otherwise indicated in the solicitation and resulting contract, the foreign acquisition restrictions in Section 225.7019 of the Defense Federal Acquisition Regulation Supplement (DFARS) apply to products described by this CID.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.

TABLE V. Lubricant, preservative, and grease fill requirements.

Code	Lubricant or preservative compound	Percentage (%) of grease fill
A	Grease in accordance with MIL-PRF-81322	Up to 25.00
B		25.01 - 50.00
C		50.01 - 80.00
D	Grease in accordance with DOD-G-24508	Up to 25.00
E		25.01 - 50.00
F		50.01 - 80.00
G	Grease in accordance with MIL-PRF-23827	Up to 25.00
H		25.01 - 50.00
J		50.01 - 80.00
K	Grease in accordance with SRI-2 or qualified equivalent	Up to 25.00
L		25.01 - 50.00
M		50.01 - 80.00
S	No fill	0.00
T	Preservation compound in accordance with MIL-DTL-197	N/A
X	Other (specify in the acquisition order)	As specified

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

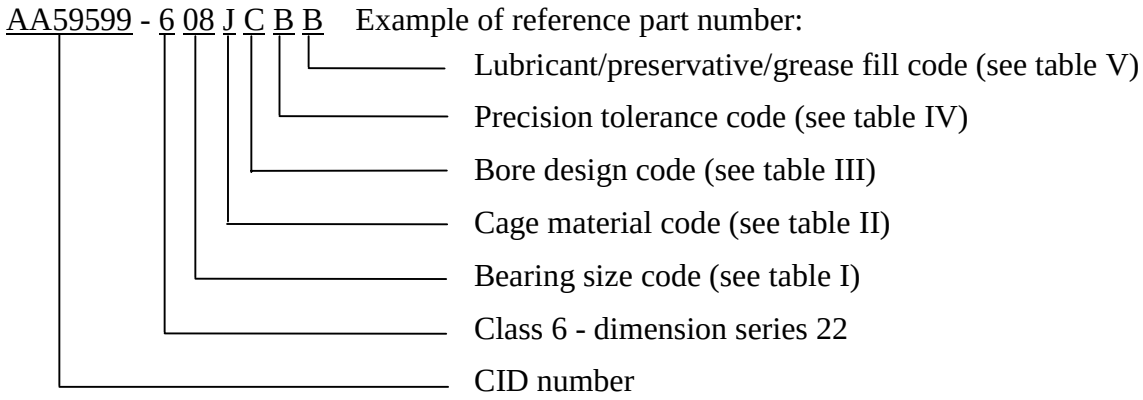
6. PACKAGING

6.1 Preservation, packing, and marking. Unless otherwise specified in the acquisition order, the bearings supplied shall be preserved, packed, and marked in accordance with MIL-DTL-197 (see 7.4(g)).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.

7. NOTES

7.1 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



AA59599 - 6 08 J C B B indicates: boundary dimension series 22; bore diameter 40 mm, outside diameter 80 mm, width 23 mm; pressed steel cage; cylindrical bore (no shields or seals); ABEC-1 tolerance class, normal radial internal clearance; filled 25-50% with grease in accordance with MIL-PRF-81322.

7.2 Sources of documents.

7.2.1 ANSI/ABMA standards. Copies of ANSI/ABMA standards may be obtained from the American Bearing Manufacturers Association, 1200 19th Street NW, Suite 300, Washington, DC 20036-2401.

7.2.2 ASTM standards. Copies of ASTM standards may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.2.3 FAR and DFARS. The FAR and DFARS may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

7.3 Sources of supply. The manufacturers and/or suppliers listed below are known to supply products that meet the salient characteristics requirements of this document. Competition is not limited to the listed firms.

FAG Bearings Corporation
Danbury, CT 06813

KOYO Corporation of U.S.A.
Westlake, OH 44145

SKF USA, Inc.
Kulpsville, PA 19443

NSK Corporation
Ann Arbor, MI 48106

NTN Bearing Corporation of America
Mt. Prospect, Il. 60056

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.

7.4 Ordering data. Acquisition documents shall specify the following information:

a. CID document number, revision, and CID PIN
b. Bearing size (with dimension/load requirements if size is unlisted) (see 3.1)
c. Cage material type (see 3.2.3)
d. Bore design (see 3.3)
e. Precision tolerance requirements (see 3.4)
f. Lubricant/preservative/grease fill requirement (see 3.6)
g. Preservation, packing, and marking requirements (see 6.1)

7.5 Codes cross-reference. Tables VI, VII, and VIII contain cross-reference data for the part identification number information as listed in FF-B-171/13 and this CID.

7.5.1 Bearing class designations. The CIDs replacing 33 of the 37 slant sheets of FF-B-171 have been assigned class codes corresponding to ABMA dimension series. Table VI lists the FF-B-171 slant sheets, the corresponding dimension series, the CID class codes, and the replacement CIDs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.

TABLE VI. Federal specification to CID cross-reference.

FF-B-171 slant sheets	ABMA dimension series	CID class codes	Replacement CID numbers
1	02, 03, 10	1, 2, 4	A-A-59581
2	19	5	A-A-59582
3	10	4	A-A-59583
4	02	1	A-A-59584
5	03	2	A-A-59585
6	04	3	A-A-59586
7	02	1	A-A-59587
8	03	2	A-A-59589
9	32	8	A-A-59595
10	33	9	A-A-59596
11	02	1	A-A-59597
12	03	2	A-A-59598
13	22	6	A-A-59599
14	23	7	A-A-59600
15	19	---	Canceled
16	10	---	Canceled
17	02	---	Canceled
18	03	---	Canceled
19	19	5	A-A-59623
20	10	4	A-A-59624
21	02	1	A-A-59625
22	03	2	A-A-59626
23	04	3	A-A-59627
24	10	4	A-A-59628
25	02	1	A-A-59629
26	03	2	A-A-59630
27	04	3	A-A-59631
28	10	4	A-A-59632
29	02	1	A-A-59633
30	03	2	A-A-59634
31	04	3	A-A-59635
32	32	8	A-A-59636
33	33	9	A-A-59637
34	32	8	A-A-59638
35	33	9	A-A-59639
36	32	8	A-A-59640
37	33	9	A-A-59641

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.

TABLE VII. Radial internal clearance and ABEC tolerance codes.

FF-B-171/13 codes			A-A-59599A codes		
Code	Radial internal clearance	Tolerance class	Code	Tolerance class	Radial internal clearance
1	Symbol 2	ABEC 1	A	ABEC 1	Symbol 2
2	Symbol 0		B		Symbol 0
3	Symbol 3		C		Symbol 3
4	Symbol 4		D		Symbol 4
			E	ABEC 3	Symbol 2
			F		Symbol 0
			G		Symbol 3
			H		Symbol 4
5	Symbol 2	ABEC-5	J	ABEC-5	Symbol 2
6	Symbol 0		K		Symbol 0
7	Symbol 3		L		Symbol 3
8	Symbol 4		M		Symbol 4
			N	ABEC-7	Symbol 2
			P		Symbol 0
			R		Symbol 3
			S		Symbol 4

TABLE VIII. Lubricant, preservative, and grease fill requirements codes.

FF-B-171/13 codes		A-A-59599A codes		
Code	Lubricant or preservative	Code	Lubricant or preservative	Grease fill %
A	Grease IAW MIL-PRF-81322	A	Grease IAW MIL-PRF-81322	Up to 25.00
B	Grease IAW DOD-G-24508	B		25.01 - 50.00
C	Grease IAW MIL-PRF-23827	C		50.01 - 80.00
D	Grease IAW SRI-2 or equivalent	D	Grease IAW DOD-G-24508	Up to 25.00
E	Preservation compound IAW MIL-DTL-197	E		25.01 - 50.00
		F		50.01 - 80.00
		G	Grease IAW MIL-PRF-23827	Up to 25.00
		H		25.01 - 50.00
		J		50.01 - 80.00
		K	Grease IAW SRI-2 or qualified equivalent	Up to 25.00
		L		25.01 - 50.00
		M		50.01 - 80.00
		S	No fill	0.00
		T	Preservation compound IAW MIL-DTL-197	N/A
		X	Other (specify in the acquisition	As specified

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.

7.6 Key words.

ABEC
cage
dynamic load rating
lubricant
precision tolerance
rings

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - OS
Air Force - 99

Reviewers:
Army - AR, EA, GL, MI
Navy - SH
Air Force - 11, 84

CIVIL AGENCY
COORDINATING ACTIVITY:
GSA - 7FXE

Preparing activity:
DLA - GS4

(Project 3110-1260)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.